Worksheet — 1.3 Exchanging Data

Name: ___ ***Date:*** ___

OCR A Level Computer Science (H446) — Component 01, Section 1.3 (Compression, encryption & hashing; Databases; Networks; Web technologies)

Time: ~60 min.

Section A — Skills practice

1. Key-term match

Match each term (A–H) to its definition (1–8). Write the letter next to the number.

	Term		Definition
A	Lossy compression	1	Translates domain names into IP addresses
B	Hashing	2	A foreign key must reference a valid existing primary key
C	RLE	3	One-way function producing a fixed-length digest
D	DNS	4	Permanently removes data; original cannot be recovered
E	Referential integrity	5	Public + private key pair used for encryption
F	Asymmetric encryption	6	Replaces runs of repeated values with value + count
G	Primary key	7	Organising data to reduce redundancy & anomalies
H	Normalisation	8	Attribute(s) that uniquely identify each record

Answers: A- **B-** C- **D-** E- **F-** G- **H-**

2. Run Length Encoding (RLE)

(a) Encode the following pixel run as RLE pairs of *(value, count)*:

```
R R R R G G B B B W W W W W
```

Working space:

(b) Decode this RLE data back into the original sequence. Each pair is *(value, count)*:

```
(3,A) (1,B) (4,A) (2,C)
```

Working space:

(c) Give **one** type of data for which RLE would *increase* the file size, and say why.

3. Symmetric vs asymmetric encryption — fill in

Complete the blanks:

a) **Symmetric** encryption uses the __ key to encrypt and decrypt. Its main weakness is the ____ problem.

b) **Asymmetric** encryption uses a __ key (shared openly) and a __ key (kept secret).

c) Data encrypted with someone's **public** key can only be decrypted with their __ key.

d) Asymmetric is __ (faster / slower) than symmetric.

e) HTTPS uses asymmetric encryption to securely exchange a __ session key, then uses __ encryption for the bulk data because it is faster.

4. Normalisation

An unnormalised `STUDENT_COURSES` table is shown below.

StudentID	StudentName	Courses	TutorID	TutorName
S01	Amy Khan	Maths, Physics	T1	Mr Bell
S02	Ben Cole	Maths	T1	Mr Bell
S03	Cara Lee	Physics, Chemistry	T2	Ms Frost

(a) State **one** repeating group that breaks **1NF**, and explain how to fix it.

(b) The `TutorName` depends on `TutorID`, not on the student. What type of dependency is this, and which normal form does it break?

(c) Redesign the data into **3NF**. Sketch the tables (table name + columns). Underline the **primary key** in each and mark any **foreign key** with (FK).

Table 1: ____________ (________)

Table 2: ____________ (________)

Table 3: ____________ (________)

Table 4 (if needed): ____________________ (_______)

5. Write the SQL

Use these tables: `Student(StudentID, Name, TutorID, YearGroup)` and `Tutor(TutorID, TutorName)`.

(a) Return the **Name** and **YearGroup** of every student in Year 12, sorted by Name (A–Z).

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(b) Return each student's **Name** alongside their **TutorName** (join the two tables on TutorID).

____

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(c) Change the **YearGroup** to 13 for the student whose StudentID is `S01`.

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6. TCP/IP layer matching

Write each protocol in the row of the layer it belongs to.

Protocols: `HTTP`, `TCP`, `IP`, `Ethernet`, `SMTP`, `UDP`, `Wi-Fi`, `FTP`

TCP/IP layer	Protocol(s)
Application	__________
Transport	__________
Network / Internet	__________
Link / Data link	__________

7. Short answer

(a) In **packet switching**, give two things that are added to each packet so it can travel the network and be reassembled.

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(b) What does **DNS** do, and why is caching used?

____

____

(c) In **PageRank**, why is a link from a high-ranked page worth more than a link from a low-ranked page?

____

(d) Give one reason why login/payment checks must run **server-side** rather than only client-side.

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8. Referential integrity vs ACID — untangle the concepts

> **Examiner note:** examiners repeatedly report that candidates confuse **Atomicity** (a property of *transactions*) with **referential integrity** (a property of the *links between tables*). This activity forces the distinction.

(a) For each statement below, write the term it describes. Choose from: **Atomicity, Consistency, Isolation, Durability, Referential integrity** (terms may be used more than once).

#	Statement	Term
1	A transaction must either complete in full or not happen at all — a half-finished transaction is rolled back.	___
2	A foreign key value must always refer to an existing primary key in the linked table.	___
3	Once a transaction has been committed, its changes survive even a power failure or crash.	___
4	Two transactions running at the same time must not interfere; the outcome is as if they ran one after the other.	___
5	A transaction must take the database from one valid state to another, obeying every rule, constraint and trigger.	___
6	An owner's record cannot be deleted while pet records still reference that owner.	___

(b) A student writes in an exam:

> *"Atomicity means that a record in one table cannot reference a record that does not exist in another table."*

Identify the concept the student has **actually** described, then write **correct** one-sentence definitions of both that concept and atomicity.

Concept actually described: ______________________________

Correct definition of it: ______________________________

Correct definition of atomicity: ______________________________

(c) Give **one** example of a real-world transaction where atomicity matters (say what the "all-or-nothing" steps are).

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9. ER diagrams & linking tables (many-to-many)

A gym stores data about **members** and the **exercise classes** they book. A member may book **many** classes; each class is booked by **many** members. Each class is run by exactly **one** instructor, and an instructor runs **many** classes.

(a) State the **degree** of the relationship between MEMBER and CLASS, and between INSTRUCTOR and CLASS.

MEMBER–CLASS: __ INSTRUCTOR–CLASS: __

(b) Explain why a many-to-many relationship **cannot** be implemented directly with a single foreign key in a relational database.

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____

(c) Design a **linking (junction) table** called `Booking` that resolves the MEMBER–CLASS relationship. Give its attributes, state its **primary key**, and mark the **foreign keys** with (FK).

`Booking( ____________________ )`

Primary key: ____________________

(d) Sketch (or describe in words) the resulting **entity-relationship diagram** for MEMBER, BOOKING, CLASS and INSTRUCTOR, labelling the degree of each relationship (1:1, 1:M or M:M).

🧠 Challenge box

A website stores user passwords. Explain why the passwords should be **hashed** rather than **encrypted**, and describe what happens when a user later logs in (refer to *one-way*, *deterministic*, and *collision*).

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Section B — Exam practice (30 marks)

Answer all questions in the spaces provided. In questions marked with an asterisk (*) the quality of your extended response will be assessed.

Q1 State what is meant by **lossless** compression, and give **one** reason why lossless (rather than lossy) compression **must** be used when compressing a program's source code file. **[2]** (AO1)

Q2 Aisha scans a black-and-white signature. One row of the scanned image contains the following 20 pixels (W = white, B = black):

```
W W W W W W W W W W B B B W W W W W W W
```

The scanner software compresses each row using run length encoding (RLE), storing the data as *(value, run length)* pairs.

(i) Write down the RLE representation of this row. **[1]** (AO2)

(ii) Uncompressed, each pixel is stored in 1 byte. Each RLE pair is stored in 2 bytes. Calculate the number of bytes **saved** by compressing this row. *Show your working.* **[2]** (AO2)

Working space:

Q3 A student writes: *"Atomicity is the database rule that every foreign key must match an existing primary key."*

Identify the concept the student has actually described, and state the **correct** meaning of atomicity. **[2]** (AO2)

Q4 A veterinary clinic uses a relational database with these tables:

`Pet(PetID, Name, Species, OwnerID)` `Owner(OwnerID, Surname, Phone)`

Write an SQL statement to display the **Name** of each pet whose Species is `Dog`, together with its owner's **Surname**, sorted in ascending order of Surname. **[4]** (AO2)

Q5 An online cinema booking system stores seat bookings in a database. Two customers, using different devices, both attempt to book the **last remaining seat** for a film at the same moment.

Explain how **isolation** and **record locking** ensure that the seat is not sold twice. **[3]** (AO2)

Q6 A user types `www.example.com` into a browser. The address is **not** stored in any local cache.

Describe how the Domain Name System (DNS) resolves this domain name into an IP address. **[3]** (AO1)

Q7 Ben enters his card details into an online shop. The site uses HTTPS, which makes use of **both** asymmetric and symmetric encryption.

Explain how asymmetric and symmetric encryption are **each** used to protect Ben's card details, and why **both** are needed. **[4]** (AO2)

Q8* A growing veterinary practice currently stores all of its data — pets, owners, appointments and payments — in a **single spreadsheet** (a flat file). The same owner's name, address and phone number are re-typed on every row for each of their pets and appointments. The practice manager is considering moving the data into a **normalised relational database**.

Discuss the benefits and drawbacks to the practice of moving from the flat file to a normalised relational database. You should consider data redundancy, data integrity and the practical impact on the practice, and give a justified recommendation. **[9]** (AO3)

Section B total: 30 marks